

Year 7 · Chapter 3 · Whole numbers: answers (teacher copy)

3.01 Rounding and estimation (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 88 000
2. Ex 1b: 87 530
3. Ex 2a: $400 + 80 + 900 + 20 = 1400$
4. Ex 2b: $900 + 300 - 400 = 800$
5. Ex 3a: $90 \times 20 = 1800$
6. Ex 3b: $2400 \div 30 = 80$

Easy · Set A (quick drill)

- | | | |
|----------|-----------|-------------|
| 1. 50 | 5. 6 000 | 9. 8 000 |
| 2. 150 | 6. 87 530 | 10. 15 000 |
| 3. 3 880 | 7. 300 | 11. 100 000 |
| 4. 1 000 | 8. 1 300 | 12. 600 |

Easy · Set B

1. 700
2. 87 500
3. 3 000
4. 90 000
5. 65 000
6. 150 000

Medium · Set C

1. $600 + 300 = 900$
2. $4000 - 2000 = 2000$
3. $40 + 70 + 400 = 510$
4. $800 - 200 + 100 = 700$
5. $7000 + 3000 = 10\,000$
6. $5000 - 4000 = 1000$

Challenging · Set D

1. $50 \times 20 = 1000$ (exact 1008)
2. $600 \times 9 = 5400$ (exact 5508)
3. $4000 \div 40 = 100$ (exact ≈ 103.9)
4. $30 \times 30 = 900$ (exact 868)

Skill drill 1 (Easy basics)

- | | | |
|------------|----------------------|---------------------|
| 1. 6 400 | 14. 260 | 27. $70 + 60 = 130$ |
| 2. 2 700 | 15. 44 000 | 28. $80 + 90 = 170$ |
| 3. 380 | 16. 970 | 29. $90 - 70 = 20$ |
| 4. 4 500 | 17. 560 | 30. $20 + 20 = 40$ |
| 5. 630 | 18. 9 000 | 31. $90 - 20 = 70$ |
| 6. 60 | 19. 480 | 32. $60 + 40 = 100$ |
| 7. 100 | 20. 5 900 | 33. $90 + 60 = 150$ |
| 8. 10 | 21. 13 000 | 34. $20 + 40 = 60$ |
| 9. 4 900 | 22. $100 + 70 = 170$ | 35. $20 + 50 = 70$ |
| 10. 4 700 | 23. $70 + 30 = 100$ | 36. $50 + 60 = 110$ |
| 11. 72 000 | 24. $20 + 90 = 110$ | 37. $20 + 50 = 70$ |
| 12. 4 800 | 25. $100 + 60 = 160$ | |
| 13. 200 | 26. $80 + 80 = 160$ | |

Skill drill 2 (Medium basics)

- | | | |
|-----------|-------------------------------|--------------------------------|
| 1. 600 | 12. 60 | 22. $50 \times 600 = 30\,000$ |
| 2. 40 000 | 13. 70 | |
| 3. 3 000 | 14. 90 000 | 23. $30 \times 200 = 6\,000$ |
| 4. 400 | 15. 20 | 24. $50 \times 1000 = 50\,000$ |
| 5. 90 000 | 16. 60 | |
| 6. 80 | 17. 4 000 | 25. $40 \times 100 = 4\,000$ |
| 7. 9 000 | 18. 5 000 | |
| 8. 50 000 | 19. 80 000 | 26. $90 \times 200 = 18\,000$ |
| 9. 80 000 | 20. 7 000 | |
| 10. 400 | 21. $80 \times 200 = 16\,000$ | 27. $20 \times 100 = 2\,000$ |
| 11. 600 | 000 | |

- | | | |
|-------------------------------|-------------------------------|--------------------------------|
| 28. $40 \times 600 = 24\,000$ | 31. $60 \times 600 = 36\,000$ | 33. $100 \times 500 = 50\,000$ |
| 29. $60 \times 800 = 48\,000$ | 32. $70 \times 600 = 42\,000$ | 34. $70 \times 50 = 3\,500$ |
| 30. $30 \times 300 = 9\,000$ | | 35. $60 \times 600 = 36\,000$ |

Extension

1. 11 500 and 12 499
2. 35 499

Exit ticket

1. Level 1: 6400
2. Level 2: $500 + 300 - 100 = 700$
3. Level 3: $4000 \div 20 = 200$

3.02 Multiplication (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 48
2. Ex 1b: 200
3. Ex 2a: 170
4. Ex 2b: 297
5. Ex 3a: 882
6. Ex 3b: 10 170

Easy · Set A (quick drill)

- | | | |
|--------|---------|-----------|
| 1. 48 | 6. 350 | 11. 132 |
| 2. 90 | 7. 56 | 12. 150 |
| 3. 180 | 8. 54 | 13. 420 |
| 4. 88 | 9. 60 | 14. 1 000 |
| 5. 100 | 10. 120 | |

Easy · Set B

1. 170
2. 200
3. 297
4. 80
5. 180
6. 231

Medium · Set C

1. 882
2. 832
3. 2 250
4. 744
5. 3 702
6. 4 131

Challenging · Set D

1. 10 170
2. 1 081
3. 14 040
4. 1 368 people

Skill drill 1 (Easy basics)

- | | | |
|---------|--------|-------------|
| 1. 42 | 12. 90 | 23. 100 |
| 2. 80 | 13. 18 | 24. 121 |
| 3. 16 | 14. 24 | 25. 32 |
| 4. 99 | 15. 64 | 26. 55 |
| 5. 80 | 16. 72 | 27. 24 |
| 6. 8 | 17. 84 | 28. 72 |
| 7. 20 | 18. 33 | 29. 66 700 |
| 8. 15 | 19. 8 | 30. 70 700 |
| 9. 30 | 20. 40 | 31. 234 000 |
| 10. 81 | 21. 24 | 32. 5 170 |
| 11. 110 | 22. 36 | 33. 44 500 |

34. 8 740	38. 4 710	42. 5 400
35. 37 000	39. 2 250	43. 502 000
36. 92 400	40. 284 000	44. 43 700
37. 600	41. 68 000	

Skill drill 2 (Medium basics)

1. 189	12. 486	23. 6 201
2. 405	13. 310	24. 2 240
3. 275	14. 364	25. 4 893
4. 126	15. 497	26. 7 830
5. 450	16. 265	27. 1 412
6. 212	17. 513	28. 1 888
7. 96	18. 392	29. 1 554
8. 475	19. 272	30. 4 210
9. 305	20. 156	31. 2 152
10. 125	21. 5 190	32. 6 136
11. 348	22. 7 528	

Extension

1. $3 (37 \times 5 = 185)$
2. e.g. 25×40 or 20×50

Exit ticket

1. Level 1: 70
2. Level 2: 1 458
3. Level 3: 3 456

3.03 Division (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 240
2. Ex 1b: 16
3. Ex 2a: 124
4. Ex 2b: 132
5. Ex 3a: 162
6. Ex 3b: 121

Easy · Set A (quick drill)

1. 240	6. 7	11. 12
2. 8	7. 8	12. 12
3. 9	8. 9	13. 16
4. 35	9. 12	14. 50
5. 12	10. 45	

Easy · Set B

1. 16
2. 18
3. 30
4. 32
5. 81
6. 12

Medium · Set C

1. 124
2. 132
3. 162
4. 121
5. 235
6. 156

Challenging · Set D

1. 79 r 1
2. 250 r 3
3. 714 r 2
4. 6 buses ($250 \div 45 = 5 \text{ r } 25$)

Skill drill 1 (Easy basics)

1. 5	3. 12	5. 8
2. 7	4. 5	6. 7

7. 9	20. 6	33. 94
8. 8	21. 2	34. 31
9. 6	22. 11	35. 38
10. 10	23. 4	36. 90
11. 2	24. 4	37. 92
12. 2	25. 9	38. 11
13. 10	26. 9	39. 64
14. 10	27. 9	40. 50
15. 6	28. 5	41. 34
16. 3	29. 31	42. 29
17. 6	30. 12	43. 52
18. 5	31. 43	44. 74
19. 3	32. 50	

Skill drill 2 (Medium basics)

1. 35	12. 34	23. 94 r 6
2. 55	13. 29	24. 189 r 2
3. 27	14. 28	25. 19
4. 47	15. 47	26. 98
5. 49	16. 38	27. 62 r 8
6. 15	17. 39	28. 36 r 3
7. 41	18. 47	29. 164 r 4
8. 56	19. 56	30. 161 r 3
9. 28	20. 31	31. 117 r 2
10. 56	21. 55 r 3	32. 147 r 4
11. 32	22. 119 r 3	

Extension

1. 26 or 38
2. 6

Exit ticket

1. Level 1: 36
2. Level 2: 123
3. Level 3: 156 r 2

3.04 Divisibility tests (booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: yes; no; no
2. Ex 1b: no (digit sum 11); no
3. Ex 2a: yes (08); yes ($208 \div 8 = 26$)
4. Ex 2b: no (not divisible by 3)
5. Ex 3a: B: 24
6. Ex 3b: 2 (522)

Easy · Set A (quick drill)

1. yes	6. yes	11. 11, no
2. no	7. 12, yes	12. 27, yes
3. yes	8. 1, no	13. 18, yes
4. yes	9. 12, yes	14. 10, no
5. no	10. 18, yes	

Easy · Set B

1. yes (12)
2. no
3. yes (18)
4. no (1)
5. yes (12)
6. yes (18)

Medium · Set C

1. by 4, 6
2. by 4, 8
3. by 4, 6
4. by 4, 8
5. by 6
6. by 4, 8

Challenging · Set D

- by 2, 3, 4, 5, 6, 8, 9, 10
- 24, 96, 120
- 1, 4, 7
- No: $9 = 3 \times 3$ so anything divisible by 9 is divisible by 3

Skill drill 1 (Easy basics)

- | | | |
|---------|------------|-------------|
| 1. no | 15. no | 29. 19, no |
| 2. yes | 16. no | 30. 13, no |
| 3. no | 17. no | 31. 20, no |
| 4. yes | 18. no | 32. 7, no |
| 5. no | 19. yes | 33. 18, yes |
| 6. no | 20. no | 34. 24, yes |
| 7. no | 21. no | 35. 13, no |
| 8. no | 22. no | 36. 24, yes |
| 9. no | 23. yes | 37. 10, no |
| 10. no | 24. no | 38. 13, no |
| 11. yes | 25. 13, no | 39. 16, no |
| 12. no | 26. 7, no | 40. 28, no |
| 13. no | 27. 22, no | |
| 14. no | 28. 29, no | |

Skill drill 2 (Medium basics)

- | | | |
|--------|---------|---------|
| 1. yes | 12. no | 23. yes |
| 2. yes | 13. yes | 24. yes |
| 3. yes | 14. no | 25. yes |
| 4. yes | 15. no | 26. yes |
| 5. no | 16. no | 27. yes |
| 6. yes | 17. yes | 28. yes |
| 7. no | 18. no | 29. yes |
| 8. no | 19. no | 30. no |
| 9. no | 20. yes | 31. no |
| 10. no | 21. no | 32. yes |
| 11. no | 22. yes | |

Extension

- Yes: ends in 0 (by 5) and digit sum 18 (by 3)
- 1020

Exit ticket

- Level 1: yes
- Level 2: yes (digit sum 18)
- Level 3: yes: even and digit sum 15

3.05 Long division (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: 26
- Ex 1b: 124
- Ex 2a: 256 r 7
- Ex 2b: 54
- Ex 3a: 156 r 8
- Ex 3b: 23 buses

Easy · Set A (quick drill)

- | | | |
|--------|---------|--------|
| 1. 84 | 6. 144 | 11. 72 |
| 2. 150 | 7. 52 | 12. 51 |
| 3. 96 | 8. 84 | 13. 88 |
| 4. 120 | 9. 69 | 14. 95 |
| 5. 96 | 10. 105 | |

Easy · Set B

- 8
- 7
- 6
- 16
- 8
- 9

Medium · Set C

- 26
- 124
- 54
- 112
- 255
- 234

Challenging · Set D

- 256 r 7
- 156 r 8
- 470 r 6
- 23 buses

Skill drill 1 (Easy basics)

- | | | |
|---------|-----------|------------|
| 1. 147 | 15. 120 | 29. 2 r 4 |
| 2. 63 | 16. 68 | 30. 9 r 15 |
| 3. 207 | 17. 39 | 31. 8 r 4 |
| 4. 32 | 18. 42 | 32. 7 r 11 |
| 5. 50 | 19. 144 | 33. 4 r 16 |
| 6. 70 | 20. 144 | 34. 5 r 4 |
| 7. 26 | 21. 108 | 35. 4 r 11 |
| 8. 105 | 22. 136 | 36. 5 r 15 |
| 9. 120 | 23. 85 | 37. 9 r 8 |
| 10. 42 | 24. 78 | 38. 5 |
| 11. 115 | 25. 3 | 39. 7 r 11 |
| 12. 120 | 26. 2 | 40. 2 r 19 |
| 13. 133 | 27. 5 r 1 | |
| 14. 65 | 28. 4 r 5 | |

Skill drill 2 (Medium basics)

- | | | |
|-------|------------|-------------|
| 1. 87 | 6. 44 | 11. 62 r 16 |
| 2. 71 | 7. 53 | 12. 58 r 21 |
| 3. 93 | 8. 22 | 13. 21 r 3 |
| 4. 97 | 9. 71 | 14. 96 r 8 |
| 5. 85 | 10. 23 r 4 | 15. 46 r 1 |

Extension

- 994
- 993

Exit ticket

- Level 1: 6
- Level 2: 108
- Level 3: 148 r 4

3.06 Roman numerals (booklet p. 33)

WE DO examples (teacher)

- Ex 1a: 37
- Ex 1b: 240
- Ex 2a: 24
- Ex 2b: XXVIII
- Ex 3a: LXXIV
- Ex 3b: CXXV

Easy · Set A (quick drill)

- | | | |
|--------|---------|----------|
| 1. 8 | 6. 15 | 11. XL |
| 2. 12 | 7. III | 12. LII |
| 3. 37 | 8. IX | 13. XCIX |
| 4. 60 | 9. XIV | 14. CL |
| 5. 210 | 10. XIX | |

Easy · Set B

- 4
- 9
- 24
- 40
- 240

6. 90

Medium · Set C

- VI
- XIII
- XXVIII
- LXXIV
- CXXV
- XLIX

Challenging · Set D

- MCMXCIX
- 2026
- e.g. 2026 = MMXXVI
- Only I before V or X is allowed (99 = XCIX)

Skill drill 1 (Easy basics)

- | | | |
|--------|------------|-------------|
| 1. 98 | 15. 85 | 29. XCVII |
| 2. 61 | 16. 82 | 30. LXXV |
| 3. 39 | 17. 13 | 31. XVII |
| 4. 18 | 18. 90 | 32. L |
| 5. 8 | 19. 17 | 33. XXVI |
| 6. 79 | 20. 81 | 34. LXXVIII |
| 7. 59 | 21. 76 | 35. XVIII |
| 8. 19 | 22. 92 | 36. LXXXII |
| 9. 86 | 23. 75 | 37. LVIII |
| 10. 23 | 24. 28 | 38. XXXIX |
| 11. 67 | 25. XCVIII | 39. XLIII |
| 12. 42 | 26. XLI | 40. XCIX |
| 13. 83 | 27. LXXXV | |
| 14. 97 | 28. LXXVII | |

Skill drill 2 (Medium basics)

- | | | |
|----------|--------------|---------------|
| 1. 787 | 12. 1301 | 23. MCMXCIII |
| 2. 619 | 13. 1448 | 24. MCMXI |
| 3. 1839 | 14. 1761 | 25. MM |
| 4. 802 | 15. 1038 | 26. MCMXCV |
| 5. 1980 | 16. 242 | 27. MCM |
| 6. 1265 | 17. MMIII | 28. MCMLXXVII |
| 7. 1186 | 18. MCMXXXV | 29. MCMLXVI |
| 8. 1215 | 19. MCMXXXII | 30. MCMXXII |
| 9. 518 | 20. MMXIII | 31. MCMXIV |
| 10. 1974 | 21. MCMXLV | |
| 11. 622 | 22. MCMLIX | |

Extension

- CCLV
- CVIII

Exit ticket

- Level 1: 16
- Level 2: XXXIX
- Level 3: MMXXIV

3.07 Index notation and powers (booklet p.

39)

WE DO examples (teacher)

- Ex 1a: 3^4
- Ex 1b: 7^7
- Ex 2a: 36
- Ex 2b: 125
- Ex 3a: 100 000
- Ex 3b: 256

Easy · Set A (quick drill)

- | | | |
|----------|----------|---------------------|
| 1. 4^3 | 3. 9^2 | 5. 10^4 |
| 2. 6^5 | 4. 2^6 | 6. $5^3 \times 3^2$ |

7. 8

8. 9

9. 25

10. 1000

11. 16

12. 32

13. 1

14. 27

Easy · Set B

- $8 \times 8 \times 8$
- $3 \times 3 \times 3 \times 3 \times 3$
- 11×11
- $2 \times 2 \times 2 \times 2$
- 7
- $2 \times 2 \times 2 \times 5 \times 5$

Medium · Set C

- 36
- 125
- 32
- 10 000
- 81
- 1

Challenging · Set D

- 256
- $64 - 27 = 37$
- $8 \times 9 = 72$
- $2^5 = 32$ is larger than $5^2 = 25$

Skill drill 1 (Easy basics)

- | | | |
|-----------|-----------|---------|
| 1. 6^2 | 11. 8^3 | 21. 8 |
| 2. 9^5 | 12. 2^3 | 22. 64 |
| 3. 5^6 | 13. 4^6 | 23. 25 |
| 4. 2^2 | 14. 3^3 | 24. 64 |
| 5. 4^2 | 15. 6^4 | 25. 243 |
| 6. 2^5 | 16. 36 | 26. 81 |
| 7. 7^3 | 17. 49 | 27. 16 |
| 8. 8^4 | 18. 27 | 28. 125 |
| 9. 9^6 | 19. 100 | 29. 9 |
| 10. 7^6 | 20. 81 | 30. 16 |
| | | 31. 32 |

Skill drill 2 (Medium basics)

- | | | |
|------------------------------|---------------------------------|---------|
| 1. base 5, index 2,
= 25 | 9. base 6, index 3,
= 216 | 17. 23 |
| 2. base 2, index 4,
= 16 | 10. base 4, index 4,
= 256 | 18. 25 |
| 3. base 3, index 2,
= 9 | 11. base 6, index 2,
= 36 | 19. 34 |
| 4. base 2, index 3,
= 8 | 12. base 7, index 3,
= 343 | 20. 25 |
| 5. base 7, index 2,
= 49 | 13. base 2, index 2,
= 4 | 21. 13 |
| 6. base 5, index 4,
= 625 | 14. base 7, index 4,
= 2 401 | 22. 80 |
| 7. base 3, index 4,
= 81 | 15. base 4, index 2,
= 16 | 23. 144 |
| 8. base 3, index 3,
= 27 | 16. base 4, index 3,
= 64 | 24. 31 |
| | | 25. 116 |
| | | 26. 500 |
| | | 27. 13 |
| | | 28. 17 |
| | | 29. 432 |
| | | 30. 141 |
| | | 31. 16 |

Extension

- $3^4 = 9^2$
- $n = 10$

Exit ticket

- Level 1: 7^3
- Level 2: 64
- Level 3: $9 \times 16 = 144$

3.08 Square roots and cube roots (booklet p.

45)

WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 12
3. Ex 2a: 20
4. Ex 2b: 9
5. Ex 3a: $\sqrt{144} = 12$
6. Ex 3b: 7 and 8

Easy · Set A (quick drill)

- | | | |
|-------|--------|--------|
| 1. 4 | 6. 1 | 11. 3 |
| 2. 7 | 7. 5 | 12. 4 |
| 3. 9 | 8. 8 | 13. 5 |
| 4. 10 | 9. 11 | 14. 10 |
| 5. 12 | 10. 20 | |

Easy · Set B

1. 2
2. 3
3. 4
4. 5
5. 10
6. 1

Medium · Set C

1. 20
2. 20
3. 9
4. 15
5. $6 + 8 = 14$
6. $6 \times 2 = 12$

Challenging · Set D

1. 9 and 10
2. 4 and 5
3. $\sqrt{100} = 10$; no ($6 + 8 = 14$)
4. 14 cm

Skill drill 1 (Easy basics)

- | | | |
|--------|---------|---------|
| 1. 2 | 13. 13 | 25. 4 |
| 2. 6 | 14. 3 | 26. 1 |
| 3. 22 | 15. 5 | 27. 9 |
| 4. 14 | 16. 19 | 28. 144 |
| 5. 20 | 17. 9 | 29. 25 |
| 6. 17 | 18. 7 | 30. 81 |
| 7. 21 | 19. 23 | 31. 100 |
| 8. 15 | 20. 16 | 32. 256 |
| 9. 12 | 21. 289 | 33. 121 |
| 10. 24 | 22. 400 | 34. 169 |
| 11. 25 | 23. 49 | 35. 196 |
| 12. 1 | 24. 36 | 36. 225 |

Skill drill 2 (Medium basics)

- | | | |
|---------|---------------|---------------|
| 1. 2 | 13. 8 | 25. 4 and 5 |
| 2. 729 | 14. 10 | 26. 12 and 13 |
| 3. 216 | 15. 4 | 27. 5 and 6 |
| 4. 125 | 16. 1 | 28. 4 and 5 |
| 5. 64 | 17. 7 | 29. 8 and 9 |
| 6. 9 | 18. 512 | 30. 9 and 10 |
| 7. 8 | 19. 6 | 31. 11 and 12 |
| 8. 343 | 20. 1 | 32. 10 and 11 |
| 9. 1000 | 21. 7 and 8 | 33. 11 and 12 |
| 10. 27 | 22. 7 and 8 | 34. 7 and 8 |
| 11. 3 | 23. 10 and 11 | 35. 12 and 13 |
| 12. 5 | 24. 12 and 13 | |

Extension

1. 70 and 30
2. 7 cm

Exit ticket

1. Level 1: 8
2. Level 2: 5
3. Level 3: $11 - 10 = 1$

3.09 Prime and composite numbers (booklet

p. 51)

WE DO examples (teacher)

1. Ex 1a: 1, 2, 3, 4, 6, 12; 1, 13
2. Ex 1b: neither
3. Ex 2a: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
4. Ex 2b: No: $51 = 3 \times 17$
5. Ex 3a: 29, 31, 37, 41, 43, 47
6. Ex 3b: 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58

Easy · Set A (quick drill)

- | | | |
|-----------------------|----------------------|-------|
| 1. 1, 2, 3, 6 | 6. 1, 2, 3, 4, 6, 9, | 10. C |
| 2. 1, 11 | 12, 18, 36 | 11. P |
| 3. 1, 3, 5, 15 | 7. P | 12. C |
| 4. 1, 2, 4, 5, 10, 20 | 8. C | 13. C |
| 5. 1, 17 | 9. P | 14. P |

Easy · Set B

1. prime
2. composite
3. prime
4. composite
5. prime
6. composite

Medium · Set C

1. 53, 59, 61, 67
2. 21, 22, 24, 25, 26, 27, 28, 30
3. 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58
4. 49, 51, 55, 57, 63, 65, 69

Challenging · Set D

1. every other even number has 2 as a factor
2. e.g. $5 + 31$, $7 + 29$, $13 + 23$, $17 + 19$
3. 3,5; 5,7; 11,13; 17,19; 29,31; 41,43
4. yes, 31

Skill drill 1 (Easy basics)

- | | | |
|-------|-------|------------------------|
| 1. C | 15. C | 29. 1, 2, 3, 4, 6, 8, |
| 2. C | 16. C | 12, 24 |
| 3. C | 17. P | 30. 1, 13 |
| 4. C | 18. C | 31. 1, 5, 25 |
| 5. C | 19. C | 32. 1, 31 |
| 6. C | 20. C | 33. 1, 41 |
| 7. C | 21. P | 34. 1, 2, 3, 4, 6, 12 |
| 8. P | 22. C | 35. 1, 5, 7, 35 |
| 9. C | 23. C | 36. 1, 3, 5, 9, 15, 45 |
| 10. C | 24. C | 37. 1, 2, 4, 11, 22, |
| 11. P | 25. P | 44 |
| 12. C | 26. C | 38. 1, 2, 23, 46 |
| 13. C | 27. C | 39. 1, 3, 9 |
| 14. P | 28. C | 40. 1, 2, 3, 6, 7, 14, |
| | | 21, 42 |

Skill drill 2 (Medium basics)

- | | | |
|------|-------|-------|
| 1. 5 | 5. 12 | 9. 2 |
| 2. 4 | 6. 2 | 10. 4 |
| 3. 2 | 7. 8 | 11. 6 |
| 4. 4 | 8. 8 | 12. 4 |

- | | | |
|------------------------|------------------------|-------------------------|
| 13. 4 | 20. e.g. 2×32 | 27. e.g. 11×11 |
| 14. 4 | 21. P | 28. P |
| 15. 3 | 22. e.g. 2×64 | 29. e.g. 2×77 |
| 16. 6 | 23. e.g. 7×19 | 30. e.g. 2×60 |
| 17. P | 24. e.g. 3×49 | 31. P |
| 18. e.g. 2×46 | 25. P | |
| 19. e.g. 3×33 | 26. e.g. 2×36 | |

Extension

- yes (not divisible by 2, 3, 5, 7, 11, 13)
- no: 13×17

Exit ticket

- Level 1: prime
- Level 2: 1, 2, 4, 7, 14, 28
- Level 3: 31, 37, 41, 43

3.10 Prime factors (booklet p. 57)

WE DO examples (teacher)

- Ex 1a: $2^2 \times 3 \times 5$
- Ex 1b: $2^3 \times 5^2$
- Ex 2a: $2^4 \times 3^2$
- Ex 2b: $2 \times 3^2 \times 5$
- Ex 3a: 2^8 , so $2^4 = 16$
- Ex 3b: $2^3 \times 3^6$, so $2 \times 3^2 = 18$

Easy · Set A

- $2^2 \times 3$
- 2×3^2
- $2 \times 3 \times 5$
- $2^3 \times 5$
- $3^2 \times 5$
- 2×5^2

Easy · Set A2 (quick drill)

- 2×3
- 3×5
- 3×7
- 5×7
- 2×11
- 3×13
- 2×13
- 7×11

Easy · Set B

- $2^3 \times 3$
- $2^2 \times 3^2$
- $2^3 \times 3^2$

Medium · Set C

- $2^2 \times 3 \times 5$
- $2^3 \times 5^2$
- $2^4 \times 3^2$
- $2^2 \times 3^2 \times 7$
- $3^3 \times 5^2$
- $2^3 \times 7^2$

Challenging · Set D

- $2^8 \rightarrow 16$
- $2^2 \times 3^4 \rightarrow 18$
- $2^3 \times 3^6 \rightarrow 18$
- $2^3 \times 7^3 \rightarrow 14$

Skill drill 1 (Easy basics)

- | | | |
|-------------------|--------------------|----------------------------|
| 1. 2×13 | 12. 2×2 | 23. $2 \times 3 \times 11$ |
| 2. 5×13 | 13. 13×13 | 24. 2×23 |
| 3. 3×13 | 14. 11×13 | 25. $2^2 \times 7$ |
| 4. 2×11 | 15. 5×5 | 26. $3^2 \times 5$ |
| 5. 7×7 | 16. 3×5 | 27. 3×19 |
| 6. 11×11 | 17. 5×11 | 28. $2^3 \times 3^2$ |
| 7. 3×3 | 18. 2×5 | 29. 3×5 |
| 8. 7×13 | 19. 3×7 | 30. 2×3^3 |
| 9. 2×7 | 20. 2×3 | 31. $2^3 \times 3$ |
| 10. 5×7 | 21. $2^2 \times 5$ | 32. $2 \times 5 \times 7$ |
| 11. 3×11 | 22. 2^3 | |

Skill drill 2 (Medium basics)

- | | | |
|-----------------------------|-----------|---------|
| 1. $2^3 \times 5 \times 7$ | 9. 19^2 | 17. 12 |
| 2. $2 \times 3^2 \times 17$ | 10. 10 | 18. 90 |
| 3. 3×113 | 11. 30 | 19. 4 |
| 4. 11×17 | 12. 40 | 20. 72 |
| 5. $2^2 \times 71$ | 13. 20 | 21. 60 |
| 6. 17×23 | 14. 2 | 22. 24 |
| 7. $2^7 \times 3$ | 15. 36 | 23. 120 |
| 8. 2×3^4 | 16. 18 | 24. 6 |
| | | 25. 360 |

Extension

- 5
- 12

Exit ticket

- Level 1: $2^2 \times 5$
- Level 2: $2^2 \times 3 \times 7$
- Level 3: $2^6 \times 3^2 \rightarrow 24$

3.11 Highest common factor (booklet p. 65)

WE DO examples (teacher)

- Ex 1a: 6
- Ex 1b: 4
- Ex 2a: 12
- Ex 2b: 17
- Ex 3a: 81
- Ex 3b: 36

Easy · Set A (quick drill)

- | | | |
|------|-------|-------|
| 1. 3 | 6. 1 | 11. 6 |
| 2. 4 | 7. 2 | 12. 7 |
| 3. 5 | 8. 3 | 13. 5 |
| 4. 7 | 9. 5 | 14. 6 |
| 5. 8 | 10. 4 | |

Easy · Set B

- 4
- 12
- 15
- 8
- 18
- 14

Medium · Set C

- 81
- 40
- 48
- 42
- 75
- 14

Challenging · Set D

- $3/7$
- 8

3. 15 cm
4. e.g. 24 and 30, or 30 and 36

Skill drill 1 (Easy basics)

- | | | |
|-------|-------|-----------------|
| 1. 15 | 13. 5 | 25. 1, 2, 4 |
| 2. 4 | 14. 3 | 26. 1, 2, 4 |
| 3. 4 | 15. 6 | 27. 1, 2, 4, 8 |
| 4. 2 | 16. 8 | 28. 1, 3 |
| 5. 7 | 17. 6 | 29. 1, 2, 4, 8 |
| 6. 5 | 18. 3 | 30. 1, 2, 5, 10 |
| 7. 6 | 19. 5 | 31. 1, 5 |
| 8. 9 | 20. 4 | 32. 1, 3, 5, 15 |
| 9. 7 | 21. 6 | 33. 1, 5 |
| 10. 5 | 22. 9 | 34. 1, 2, 5, 10 |
| 11. 7 | 23. 4 | 35. 1, 2, 3, 6 |
| 12. 6 | 24. 8 | 36. 1, 2 |

Skill drill 2 (Medium basics)

- | | | |
|--------|--------|--------|
| 1. 36 | 12. 7 | 23. 12 |
| 2. 9 | 13. 7 | 24. 8 |
| 3. 6 | 14. 24 | 25. 8 |
| 4. 8 | 15. 60 | 26. 10 |
| 5. 56 | 16. 32 | 27. 11 |
| 6. 10 | 17. 8 | 28. 7 |
| 7. 18 | 18. 2 | 29. 4 |
| 8. 18 | 19. 9 | 30. 10 |
| 9. 16 | 20. 10 | 31. 11 |
| 10. 11 | 21. 2 | |
| 11. 4 | 22. 5 | |

Extension

1. 24
2. 15

Exit ticket

1. Level 1: 4
2. Level 2: 14
3. Level 3: 84

3.12 Lowest common multiple (booklet p. 71)

WE DO examples (teacher)

1. Ex 1a: 12
2. Ex 1b: 24
3. Ex 2a: 90
4. Ex 2b: 35
5. Ex 3a: 400
6. Ex 3b: 180

Easy · Set A (quick drill)

- | | | |
|------------------------|-----------------------|--------|
| 1. 3, 6, 9, 12, 15, 18 | 5. 9, 18, 27, 36, 45, | 9. 20 |
| 2. 5, 10, 15, 20, 25, | 54 | 10. 18 |
| 30 | 6. 15, 30, 45, 60, | 11. 40 |
| 3. 8, 16, 24, 32, 40, | 75, 90 | 12. 30 |
| 48 | 7. 10 | 13. 24 |
| 4. 12, 24, 36, 48, | 8. 12 | 14. 36 |
| 60, 72 | | |

Easy · Set B

1. 6
2. 20
3. 18
4. 21
5. 24
6. 30

Medium · Set C

1. 90
2. 400
3. 180
4. 42
5. 72
6. 180

Challenging · Set D

1. 7:00 am (60 minutes)
2. 24
3. 4 packs of hot dogs, 5 packs of buns (40 each)
4. 3, 6, 12, 24

Skill drill 1 (Easy basics)

- | | | |
|--------|---------|-----------------|
| 1. 72 | 13. 60 | 25. 8, 16, 24 |
| 2. 12 | 14. 18 | 26. 15, 30, 45 |
| 3. 56 | 15. 30 | 27. 6, 12, 18 |
| 4. 22 | 16. 42 | 28. 42, 84, 126 |
| 5. 9 | 17. 40 | 29. 21, 42, 63 |
| 6. 10 | 18. 45 | 30. 24, 48, 72 |
| 7. 6 | 19. 35 | 31. 10, 20, 30 |
| 8. 60 | 20. 12 | 32. 4, 8, 12 |
| 9. 10 | 21. 55 | 33. 8, 16, 24 |
| 10. 56 | 22. 60 | 34. 20, 40, 60 |
| 11. 15 | 23. 132 | 35. 21, 42, 63 |
| 12. 12 | 24. 24 | 36. 6, 12, 18 |

Skill drill 2 (Medium basics)

- | | | |
|---------|---------|---------|
| 1. 20 | 12. 60 | 23. 72 |
| 2. 45 | 13. 312 | 24. 84 |
| 3. 66 | 14. 12 | 25. 70 |
| 4. 255 | 15. 221 | 26. 30 |
| 5. 132 | 16. 90 | 27. 72 |
| 6. 14 | 17. 24 | 28. 8 |
| 7. 40 | 18. 140 | 29. 105 |
| 8. 357 | 19. 360 | 30. 90 |
| 9. 105 | 20. 60 | 31. 180 |
| 10. 56 | 21. 20 | |
| 11. 399 | 22. 10 | |

Extension

1. 72
2. 60 seconds

Exit ticket

1. Level 1: 12
2. Level 2: 48
3. Level 3: 84